Irrigation and Agricultural Water Management

Agriculture RAG Knowledge Base • Educational reference document

Irrigation management is the planned application of water to crops when rainfall does not adequately meet crop demand. The objective is not simply to apply more water, but to deliver enough water at the appropriate time while minimizing losses.

Common irrigation systems include surface irrigation, sprinkler irrigation, and drip irrigation. Drip systems deliver water near the crop root zone and can reduce unnecessary wetting when properly designed and maintained. Sprinklers distribute water through pressurized systems and can be useful for many crops, but wind and evaporation can affect efficiency.

Irrigation scheduling should consider crop growth stage, soil characteristics, weather, rainfall, and available soil moisture. Applying water too early or too frequently can cause waterlogging and nutrient losses. Applying too little water can cause crop stress and yield reduction.

Water-use efficiency can be improved through appropriate scheduling, maintenance of irrigation equipment, soil-moisture monitoring, mulching, rainwater harvesting, and improved field distribution. Fertigation can combine irrigation and nutrient application, but fertilizer injection must be managed carefully to avoid uneven distribution or contamination of water sources.

For rainfed agriculture, useful approaches include conserving soil moisture, harvesting rainfall, reducing runoff, selecting suitable crop varieties, and aligning crop management with expected water availability.

Reference basis

• FAO, Irrigation Management

• FAO, Rainfed Agriculture

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.